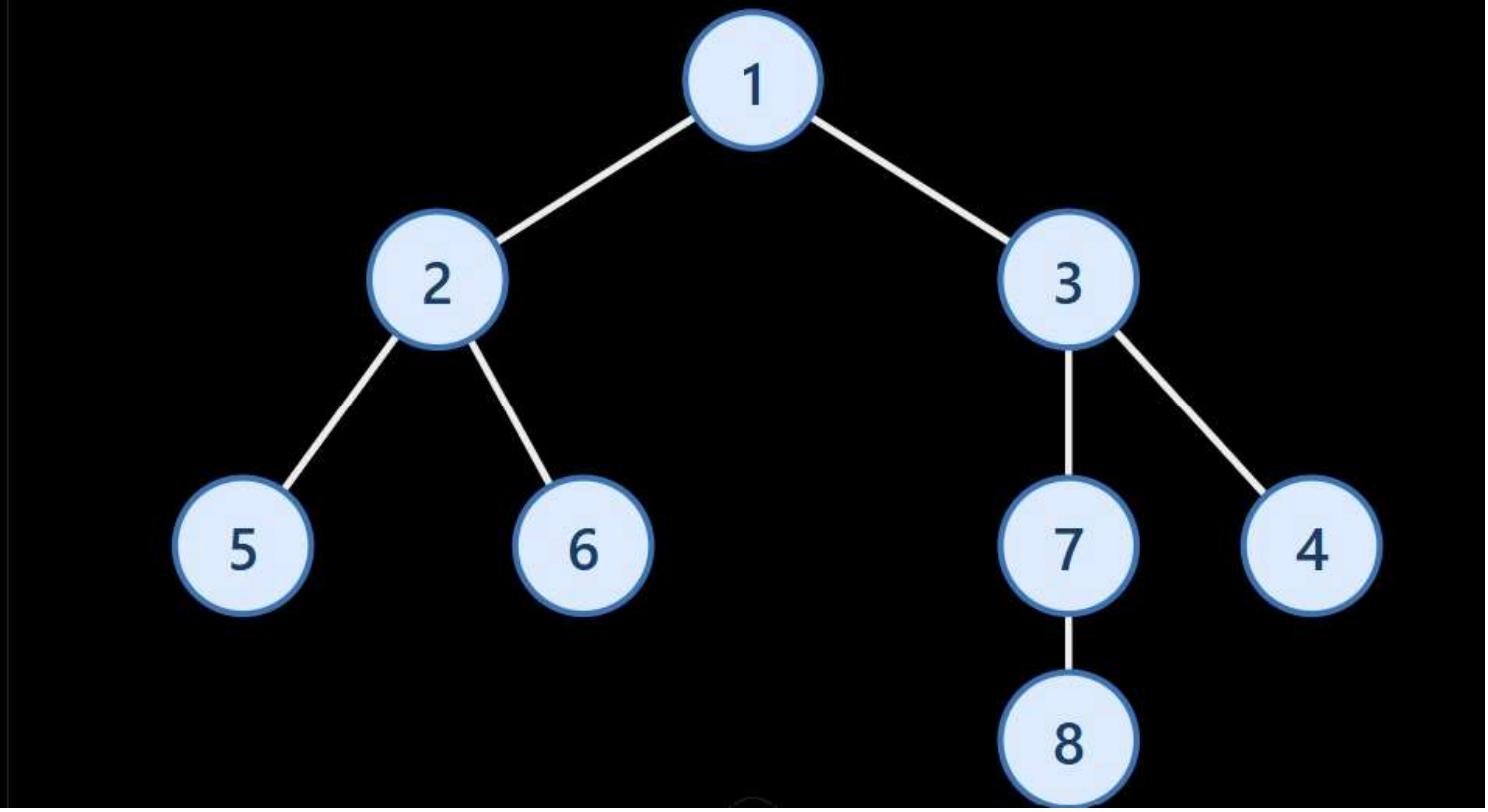
Question 3: BFS and DFS

Diagram



Programiz Python Online Compiler

main.py

Visualize Run ▶

```
from collections import deque

graph = {
    1: [2, 3],
    2: [5, 6],
    3: [7, 4],
    4: [3, 8],
    5: [2],
    6: [2],
    7: [3, 8],
    8: [7, 4]
}

def bfs(start):
    visited = {start}
    queue = deque([start])
    while queue:
        node = queue.popleft()
        print(node, end=" ")
        for neighbour in graph[node]:
            if neighbour not in visited:
                visited.add(neighbour)
                queue.append(neighbour)
```

Output

```
BFS:
1 2 3 5 6 7 4 8
DFS:
1 2 5 6 3 7 8 4
```

```
def dfs(node, visited):
    visited.add(node)
    print(node, end=" ")
    for neighbour in graph[node]:
        if neighbour not in visited:
            dfs(neighbour, visited)

print("BFS:")
bfs(1)

print("\nDFS:")
dfs(1, set())
print()
```